

## AIM:

## Bubble Sort with Big-O Analysis

```
1 #include <stdio.h>
2 int main() {
3     int a[50], n, i, j, temp;
4     printf("Enter number of elements: ");
5     scanf("%d", &n);
6     printf("Enter elements:\n");
7     for(i = 0; i < n; i++)
8         scanf("%d", &a[i]);
9     for(i = 0; i < n - 1; i++) {
10         for(j = 0; j < n - i - 1; j++) {
11             if(a[j] > a[j + 1]) {
12                 temp = a[j];
13                 a[j] = a[j + 1];
14                 a[j + 1] = temp;
15             }
16         }
17     }
18     printf("Sorted Array: ");
19     for(i = 0; i < n; i++)
20         printf("%d ", a[i]);
21     return 0;
22 }
```

Output

Enter number of elements: 5  
Enter elements:  
34 45 65 73 82  
Sorted Array: 34 45 65 73 82

## Result:

Bubble Sort with Big-O Analysis has been executed successfully